Hoofstuk 3

Lesing 3.2

Recap: → Doelwit: Los op vir V of I.

→ Metode:

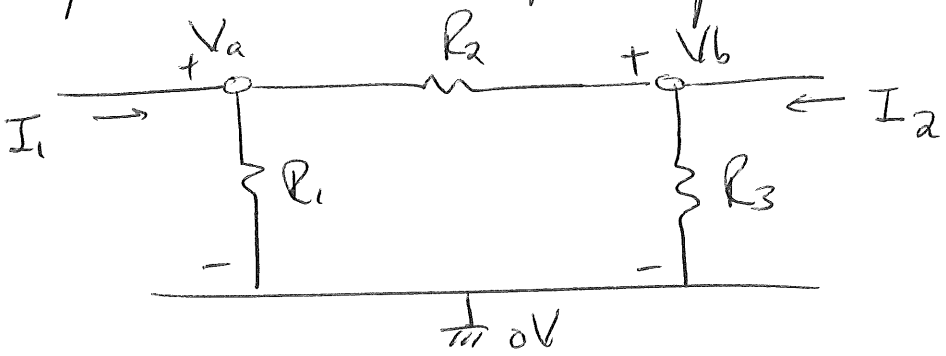
① Ω-wet ⇒ $V = \pm IR$

② KVL ⇒ $\sum V = 0$

③ KCL ⇒ $\sum I_{UIT} = \sum I_{IN}$

↳ Node analise.

* Skryf strome i.t.v. spannings en weerstande.



① $I_1 = \frac{V_a}{R_1} + \frac{V_a - V_b}{R_2}.$

② $I_2 = \frac{V_b}{R_3} + \frac{V_b - V_a}{R_2}$

$$\boxed{i = \frac{V_{FROM} - V_{TO}}{R}}$$

Stappe:

① Kies 0V.

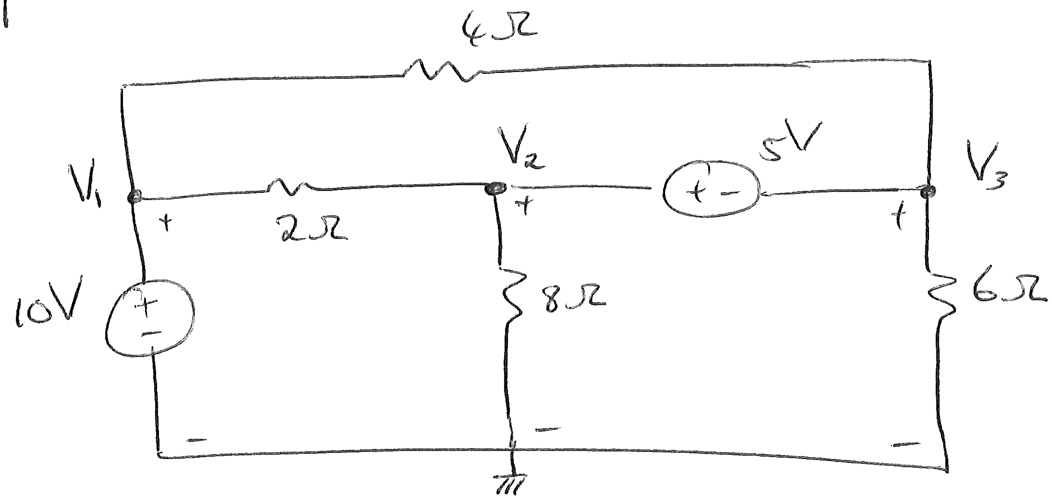
② Definieer node-spannings → ⊕ by node. ⊖ by 0V.

③ Stel vergelykings op.

④ Los op mbv: → * Substitusie → * Cramer se Wet.

* Supernodusse

Vb



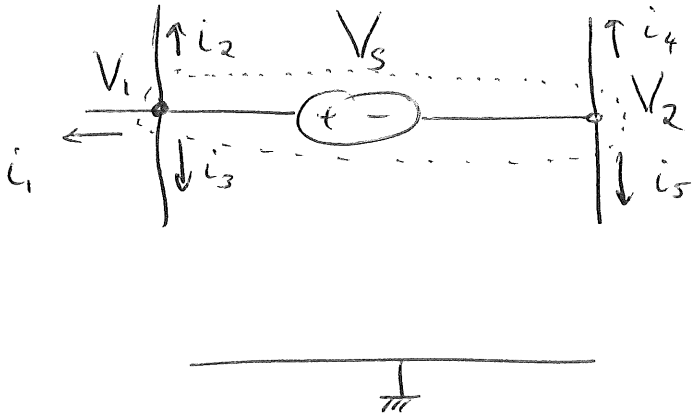
1 → $V_1 = ?$

2 → Hoe word KCL by V_2 en V_3 gedoen?

(Ω-Wet nie van toepassing op bronne nie.)

Supernode: • vorm geslote grens rondom V-bron en enige elemente || tot bron; ...

• Geslote grens word nuwe "node."



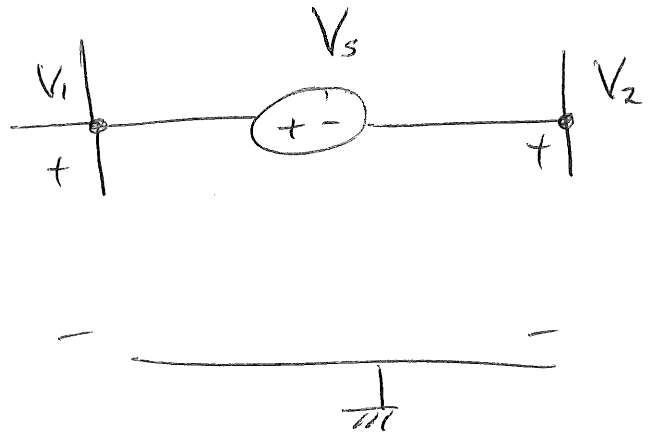
• KCL:

$i_1 + i_2 + i_3 + i_4 + i_5 = 0$

- KCL word by supernode gedoen.
- Nog 'n vergelyking word benodig.

  (Node analise sou gewoonlik 2 vgl's vir 2 nodusse gee.)

  Doen KVL rondom bron in supernode.



$$-V_1 + V_s + V_2 = 0$$

$$\therefore V_s = V_1 - V_2.$$

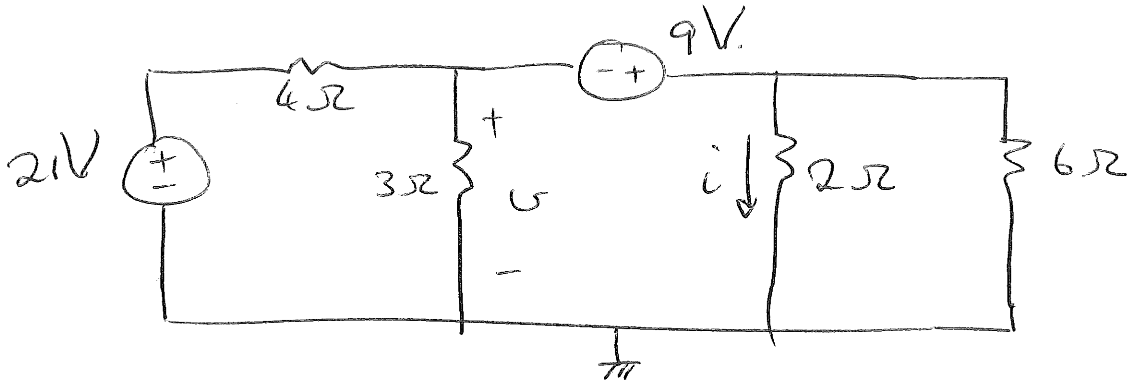
- Node-analise se vgl's vir voorbeeld.

① $V_1 = 10V$

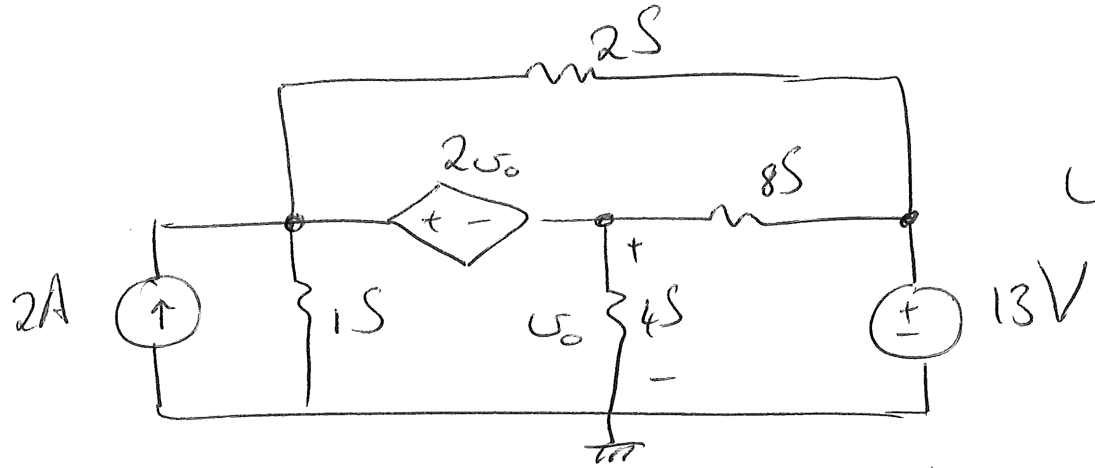
KVL: ② $V_s = V_1 - V_2.$

KCL: ③ $$\frac{V_2 - V_1}{2\Omega} + \frac{V_2}{8\Omega} + \frac{V_3}{6\Omega} + \frac{V_3 - V_1}{4\Omega} = 0$$

Vb $v = ?$ $i = ?$



Vb



$v_o = ?$